

Seeing

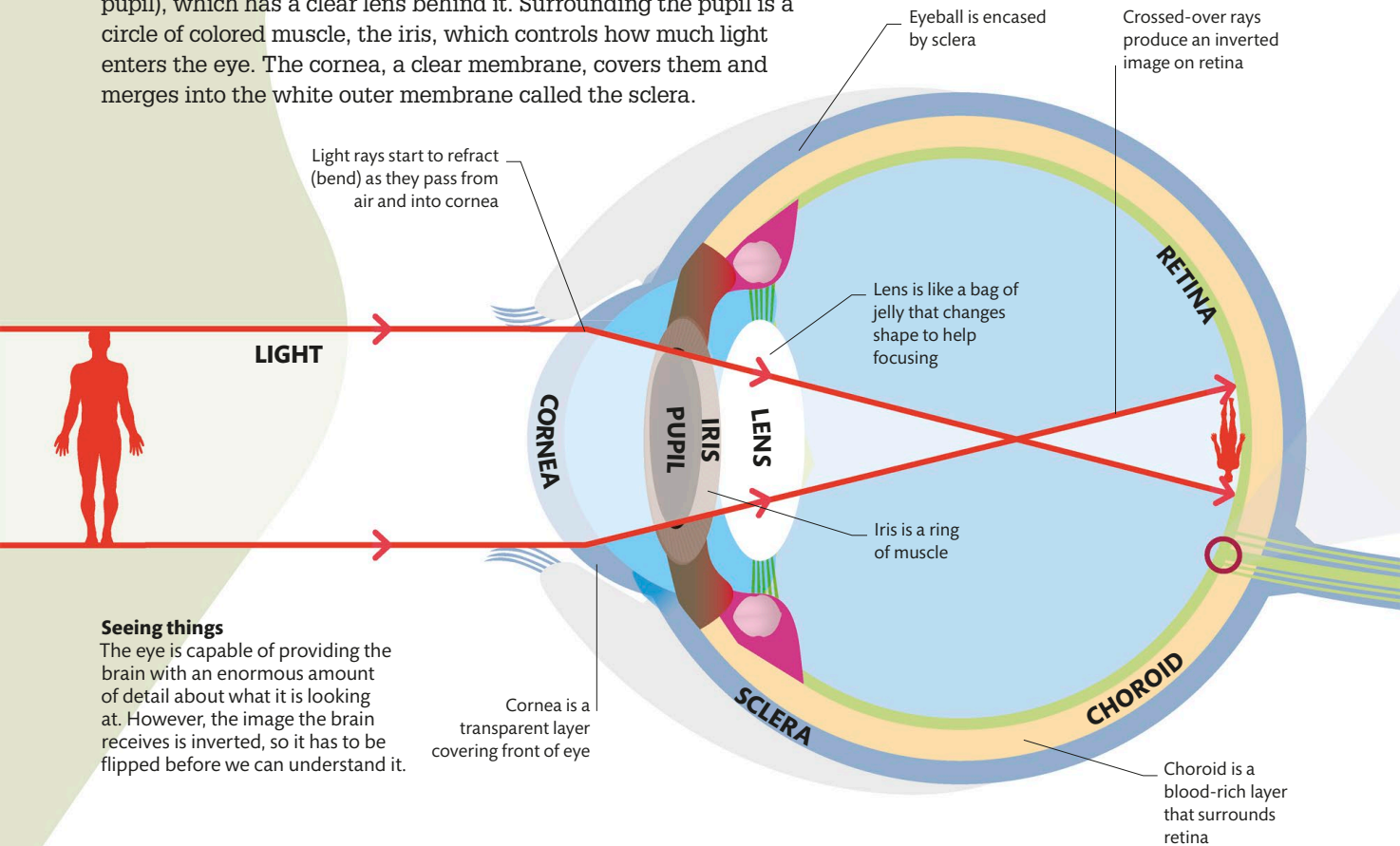
The eye provides us with probably the most important of our five senses. It gathers the light reflected by an object and delivers this information to the brain via the optic nerve.

The structure of the eye

The eyeball is roughly 1 in (2.5 cm) in diameter. At the back of the eye is the retina, which contains light-sensitive cells that connect via neurons to the optic nerve. The space inside the eyeball is filled with a jellylike substance. The front of the eye contains a hole (the pupil), which has a clear lens behind it. Surrounding the pupil is a circle of colored muscle, the iris, which controls how much light enters the eye. The cornea, a clear membrane, covers them and merges into the white outer membrane called the sclera.

WHY DO MY EYES CLOSE WHEN I SNEEZE?

When a nasal irritant triggers the brain stem control center, it causes widespread muscle contractions, including those in the eyelids. This makes you blink momentarily.



Seeing things

The eye is capable of providing the brain with an enormous amount of detail about what it is looking at. However, the image the brain receives is inverted, so it has to be flipped before we can understand it.

1

Light enters the eye

Light passes through the cornea and into the eye through the pupil. The pupil is surrounded by a ring of colored muscle, the iris, which can make the pupil contract or dilate to vary the amount of light entering.

2

Lens and focusing

Behind the iris is the lens, where the light rays are bent so the image forms on the retina. The lens is connected to muscles that allow it to change shape—it flattens for distant objects and thickens for close objects.